The dynamics of the system is given in (Analytic LQR Design for Spacecraft Control System Based on Quaternion Model) equations 2-6, which I copy below.

$$J\dot{\omega} = -\omega^{\times}(J\omega) + u \tag{1}$$

$$\omega^{\times} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \tag{2}$$

J is the inertia matrix, ω is the angular velocity in body frame.

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} f(q) & -q_3 & q_2 \\ q_3 & f(q) & -q_1 \\ -q_2 & q_1 & f(q) \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \frac{1}{2}\Omega\omega = g(q_1, q_2, q_3, \omega). \tag{3}$$

$$f(q) := q_0 = \sqrt{1 - q_1^2 - q_2^2 - q_3^2}. \tag{4}$$

The linearized system about $(q, \omega) = (\mathbf{q}_0, \boldsymbol{\omega}_0)$ is:

$$\delta\dot{\mathbf{q}} = \left.\frac{\partial g}{\partial \boldsymbol{\omega}}\right|_{(\mathbf{q}_0, \boldsymbol{\omega}_0)} \delta\boldsymbol{\omega} + \left.\frac{\partial g}{\partial \mathbf{q}}\right|_{(\mathbf{q}_0, \boldsymbol{\omega}_0)} \delta\mathbf{q}. \tag{5}$$

Where $\delta\mathbf{q} = \mathbf{q} - \mathbf{q}_0, \delta\omega = \omega - \bar{\omega}$, and :

$$\frac{\partial g}{\partial \omega} = 1/2 \begin{bmatrix} f(q) & -q_3 & q_2 \\ q_3 & f(q) & -q_1 \\ -q_2 & q_1 & f(q) \end{bmatrix} \tag{6}$$

$$\frac{\partial g}{\partial q} = 1/2 \begin{bmatrix} \frac{\partial f}{\partial q_1}\omega_1 & \frac{\partial f}{\partial q_2}\omega_1 + \omega_3 & \frac{\partial f}{\partial q_3}\omega_1 - \omega_2 \\ \frac{\partial f}{\partial q_1}\omega_2 - \omega_3 & \frac{\partial f}{\partial q_2}\omega_2 & \frac{\partial f}{\partial q_3}\omega_2 + \omega_1 \\ \frac{\partial f}{\partial q_1}\omega_3 + \omega_2 & \frac{\partial f}{\partial q_2}\omega_3 - \omega_1 & \frac{\partial f}{\partial q_3}\omega_3 \end{bmatrix} \tag{7}$$

The linearization of ω is:

$$\delta\dot{\boldsymbol{\omega}} = J^{-1}\left((J\boldsymbol{\omega}_0)^{\times} - \boldsymbol{\omega}_0^{\times} J\right)\delta\boldsymbol{\omega} + J^{-1}\mathbf{u}. \tag{8}$$

Putting them together:

$$\begin{bmatrix} \dot{\delta\boldsymbol{q}} \\ \dot{\delta\boldsymbol{\omega}} \end{bmatrix} = \begin{bmatrix} \frac{\partial g}{\partial q} & \frac{\partial g}{\partial \omega} \\ 0 & J^{-1}((J\boldsymbol{\omega}_0)^{\times} - \boldsymbol{\omega}_0^{\times} J) \end{bmatrix} \begin{bmatrix} \delta\boldsymbol{q} \\ \delta\boldsymbol{\omega} \end{bmatrix} + \begin{bmatrix} 0 \\ J^{-1} \end{bmatrix} u \tag{9}$$